DSCI 4350 · GROUP 5 · CHECKPOINT 2

Exploratory Data Analysis

OASIS-2 visit pairs, the clinical notes, and the scans that belong to them. A reading of the file before any model is trained.

Each row is one person at one visit, and a question about the next one: will the Clinical Dementia Rating be higher when they come back? It is, for 31 of 223 pairs. The pages after this say where that signal sits, and where the pictures still have nothing plain to add.

150	223	1,367	20	13.9%
patients, ages 60 to 96	visit pairs in the modeling file	T1 scans, 373 sessions	features a model may see	of pairs declined

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`oasis2_visit_pairs.csv · SHA-256 8668b506a03b3181… · random_state 42`

THE QUESTION

Will the next visit be worse?

> *Everything a model is allowed to see comes from the earlier visit. The label comes from the later one.*

Most work in this area sorts people by the dementia they already have. The score would be sitting in the same row as the features. We are asking something narrower. Standing at visit t, with the clinical measures and the scan from that day, can we tell whether CDR will have risen by visit t+1?

A row is one person moving from one visit to the next. Someone with more visits contributes more pairs. The last visit has no “next,” so it drops out, and 373 visits become 223 pairs. Stable means CDR held or, in five pairs, actually fell. Declined means it rose. We treat a half-point step the same as a larger one.

On the charts, blue is stable and clay is declined. A training AUC uses training patients only. A value of 0.50 is a coin toss. Means are computed on all 223 pairs. The test set is for the end, and it holds six declines.

A SHORT GLOSSARY

Visit pair	One row. Features from visit t, label from visit t+1.
CDR	Clinical Dementia Rating. 0 none, 0.5 very mild, 1 mild, 2 moderate.
MMSE	Mini-Mental State Examination, 0 to 30. Lower is worse.
nWBV	Normalized whole-brain volume, the brain’s share of the head.
eTIV / ASF	Head size, and the atlas scaling factor, which is about 1 / eTIV.
SES	Socioeconomic status. 1 is the highest, 5 the lowest.
CSF proxy	Share of dark, fluid-like voxels in the middle of the raw scan.
SNR	Signal-to-noise of a scan. A quality measure, not a feature.

03 · Sources
How a visit and a scan became one row.

04 · Target
31 rises in CDR, mostly a single step.

05–08 · Features
What separates, and what repeats.

09–11 · Scans
Format, four examples, quality.

12 · Models
What the three approaches should hear.

13–15 · Card
Fields, rebuild steps, limits.

FROM THE STUDY TO THE SPREADSHEET

Two archives, joined on the scan ID

The clinical table and the pictures arrive separately. `mri_id` is the stitch. Both builds are deterministic, and the test patients never get a vote in the imputation.

CLINICAL TABLE · MAKE_DATASET.PY

1. **OASIS-2, Washington University.** Longitudinal MRI and clinical visits from the Knight ADRC. Marcus et al., 2010.
2. **Kaggle copy, used as-is.** 373 visits, 15 columns, 150 patients, two to five visits each.
3. **Fill and look backward.** Two missing MMSE values, one patient, filled from that person's previous score. Change and yearly rate use only the visit before this one.
4. **Pair t with t+1.** 223 pairs. Declined is 1 when CDR rises. Then attach the MRI session measures. All 223 pairs match.
5. **Split, then impute.** About one patient in five is held out, stratified on the target. Training is split again into five grouped folds. Missing SES, 8 patients and 11 pairs, is filled with the training mode, SES = 2.

RAW SCANS · MAKE_MRI_DATASET.PY

1. **Two tar archives, 19.3 GB.** Streamed once. Nothing is unpacked to disk, which would have been about 45 GB.
2. **1,367 T1 volumes, 373 sessions.** Every scan is decoded in memory. We keep intensity, SNR, and a CSF proxy.
3. **Repeats are averaged per session.** Agreement between the first scan and the repeats is the motion check. Nineteen sessions are flagged.
4. **Every visit has a session.** 373 of 373. Every pair has scans at both ends. 223 of 223.

Two raw columns never became features. Group is assigned from a person's whole history, so it can see the outcome. Hand never varies: everyone is right-handed. Sex was recoded to `sex_male`, and names went to snake_case.

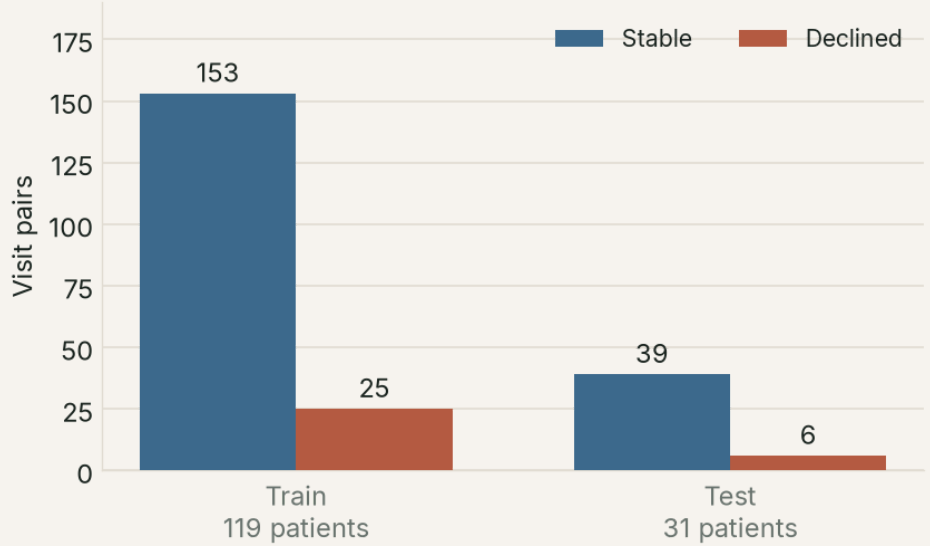
FILE	WHERE IT CAME FROM	WHAT IT HOLDS	SHA-256 BEGINS
`oasis_longitudinal.csv`	Kaggle, OASIS-2	373 visits · 28 KB	`d2f0a15ff35f`
`OAS2_RAW_PART1.tar.gz`	oasis-brains.org	771 scans · 10.9 GB · 209 sessions	`6df8fa570698`
`OAS2_RAW_PART2.tar.gz`	oasis-brains.org	596 scans · 8.4 GB · 164 sessions	`50bc39b6d911`
`oasis2_visit_pairs.csv`	This project	223 pairs · 42 KB · 20 features	`8668b506a03b`

Hashes of the raw files and of this output live in the metadata JSON

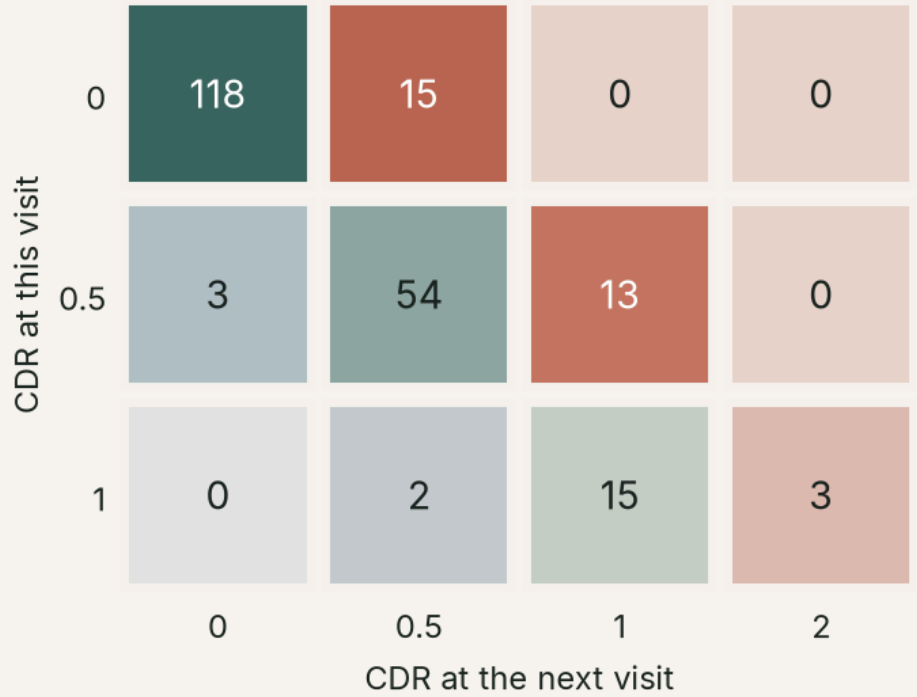
STABLE OR DECLINED

Decline is uncommon, and usually one step

Of 223 pairs, 192 stayed put and 31 moved up. The grid is that sentence with the numbers filled in. Clay cells are a rise, which is our positive class. Green cells stayed. Gray cells improved, and we still call those stable.



The patient-level split keeps the rate near 14% on both sides: 14.0% in train, 13.3% in test. No person sits in both. The harder fact is the test set. Six declines can be reported. They cannot carry a precise claim about a model. Model choice belongs on the five grouped folds.



Above the diagonal: 15 + 13 + 3 = 31. Below it, five pairs improved. Nobody jumped from CDR 0 straight to 1 or 2. Only three pairs in the whole file reach CDR 2.

Stayed Rose — declined Improved — still labeled stable

118 stayed well. CDR 0 to CDR 0. Another 15 people took the first step, from 0 to 0.5.

The middle is where it moves. Of 70 pairs already at 0.5, 54 stayed, 13 went to 1, and 3 went back to 0.

CDR 2 is three people. All of them came from CDR 1. Fifteen pairs at CDR 1 simply stay there.

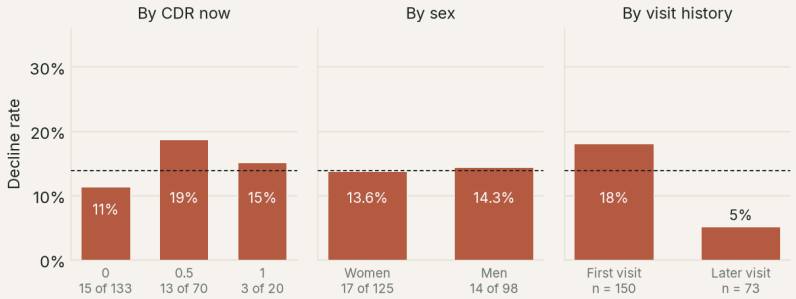
declined = 1 when CDR at t+1 is higher than CDR at t · no one declined twice

WHO THE TWO GROUPS ARE

On the averages, they look alike

Means on all 223 pairs. The marked rows are the ones that actually move. The dashed line on the rates is 13.9%, the rate for everyone.

AT THE EARLIER VISIT	STABLE · 192	DECLINED · 31
MMSE, 0–30	27.9	26.9
Days since the first visit	308	80
Has a visit before this one	36%	13%
CDR at this visit	0.24	0.31
Days until the next visit	709	780
Whole-brain volume	0.735	0.729
CSF proxy, from the scan	0.261	0.255
Age, years	76.0	76.6
Education, years	14.6	14.7



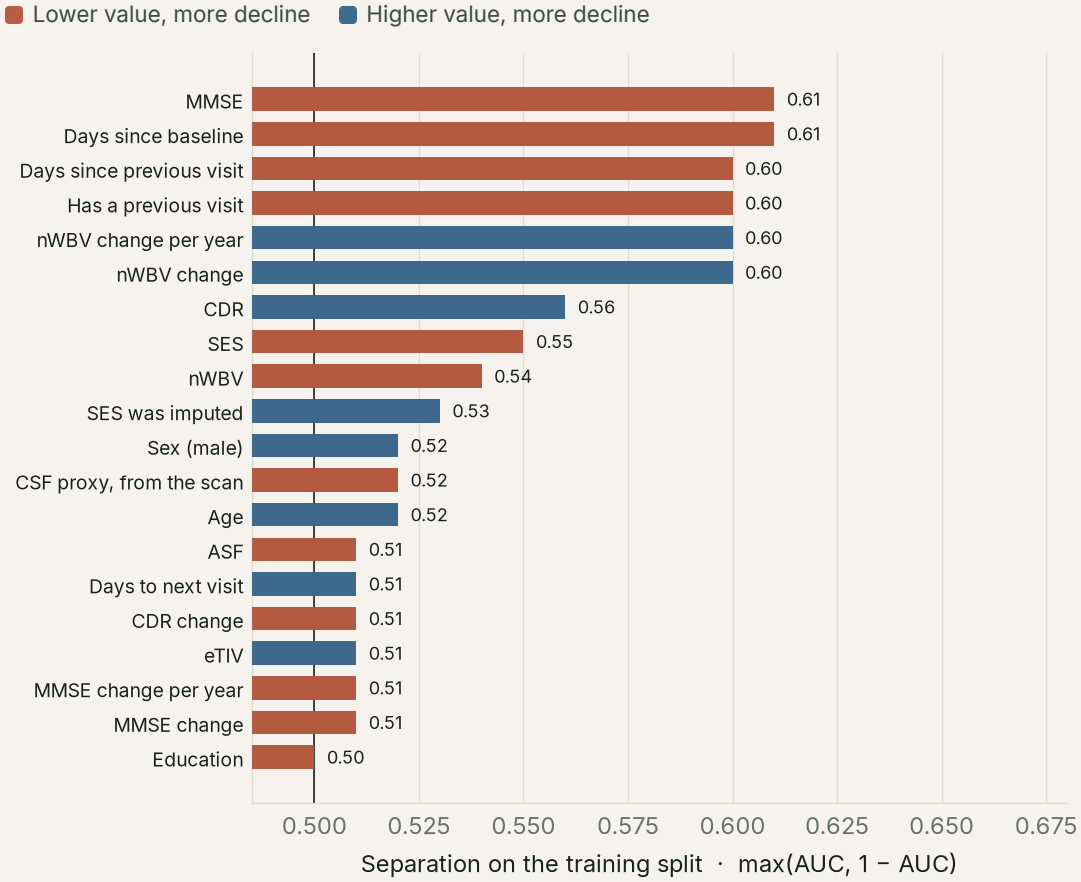
CDR 0 declines at 11% (15 of 133). CDR 0.5 declines at 19% (13 of 70). CDR 1 is 15%, and that bar is 3 of 20. Women and men sit on the same line, 13.6% and 14.3%. About 18% of first-visit pairs declined, against about 5% of later ones.

A lot of zeros are "no history yet." 150 of the 223 pairs are someone's first visit, so the change features are 0 there by construction. MMSE also crowds the ceiling at 30 in both groups. The one-point gap between 27.9 and 26.9 sits inside a standard deviation of about 3. It is still the clearest clinical difference in the file. SES leans a little, higher status a bit more decline, but SES 5 is four pairs and education has no separation at all, so we hold that lightly.

Sex counts come from the class means: 17 of 125 women, 14 of 98 men

ONE COLUMN AT A TIME

Nothing here is a strong classifier



Each bar is how well that one column, used alone, separates the classes on the training patients. We fold the AUC so 0.61 means a little separation, whichever direction it points.

The top is shared by MMSE and days since baseline, both pointing the same way: lower, more decline. Close behind are the other clocks, whether a previous visit exists, and the two nWBV change measures. Those last two point the other way. A higher value, less volume loss or a flat zero, lines up with later decline. That is the first-visit pattern again. People we have not seen before have a change of zero, and they are also the people who decline more often here.

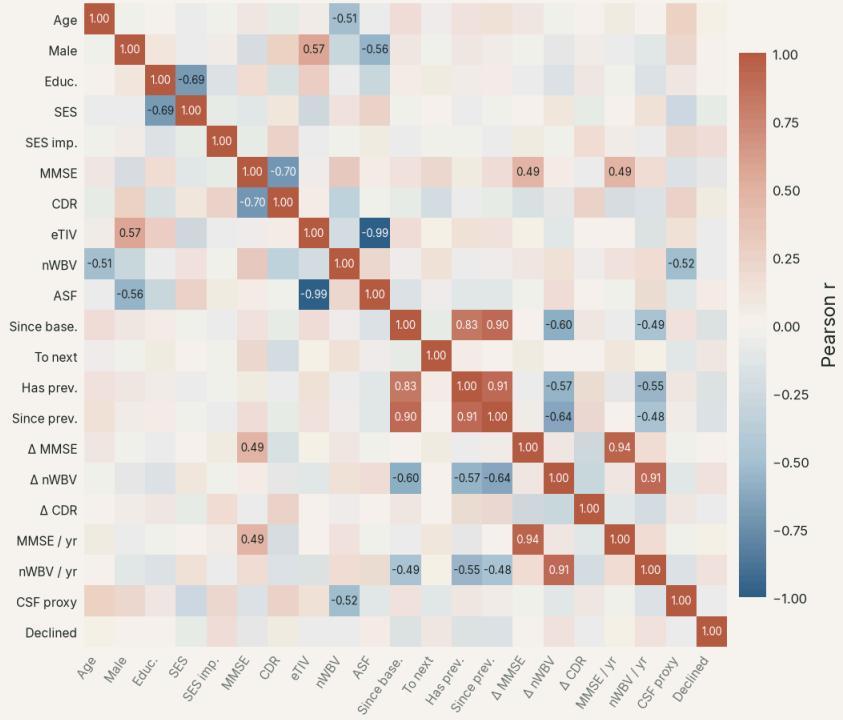
The new scan measure sits near the bottom. The CSF proxy reaches 0.52, in the same neighborhood as age and sex. Brain volume from the table is 0.54. Baseline CDR is a modest 0.56. Education is a coin toss.

max(AUC, 1 − AUC) · training patients only · the scan measure is marked in the list

PEARSON R ON ALL 223 PAIRS

A few columns are the same fact twice

Warm cells move together, cool cells move apart. Numbers are printed only where |r| is at least 0.48, so the pairs that matter can be read. The label, on the bottom row, stays pale.



Keep one head-size column

eTIV and ASF correlate at −0.99. The atlas factor is, for practical purposes, one over head size. A linear model should see only one of them.

A family of clocks

Days since baseline, days since the previous visit, and the flag for having a previous visit correlate between 0.83 and 0.91. Change and the yearly rate are near-duplicates: 0.94 for MMSE, 0.91 for nWBV.

The scan agrees with the table

The CSF proxy tracks nWBV at −0.52, and it rises with CDR (0.24). It barely touches the label (−0.05). MMSE and CDR themselves correlate at −0.70. Education and SES at −0.69, in the direction the coding implies. nWBV falls as age rises (−0.51), so volume and age belong in a model together. Men have larger heads (0.57 with eTIV).

The strongest correlation with declined is about −0.17, with the follow-up columns. Correlation is a blunt tool for a 14% outcome. The AUC ranking is the fairer picture of separation, and both pictures agree on the main point.

CSF proxy versus nWBV is −0.54 when measured on all 373 sessions

THE FULL TABLE

Every model input, with the number beside it

Means and standard deviations on all 223 pairs. The last column is the raw training AUC, so a value under 0.50 means the lower scores are the ones that decline. MMSE at 0.39 is the same result as the 0.61 on the bar chart.

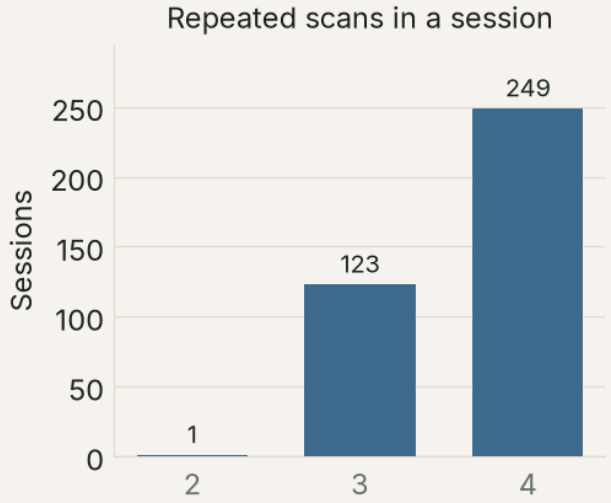
GROUP	FEATURE	RANGE	STABLE MEAN (SD)	DECLINED MEAN (SD)	TRAIN AUC
Demographics	age	60 to 96	76 (7.55)	76.6 (7.27)	0.52
	sex_male	0 to 1	0.438 (0.497)	0.452 (0.506)	0.52
	educ	6 to 23	14.6 (2.94)	14.7 (2.57)	0.50
	ses	1 to 5	2.45 (1.12)	2.19 (1.05)	0.45
	ses_imputed	0 to 1	0.0365 (0.188)	0.129 (0.341)	0.53
Clinical	mmse	7 to 30	27.9 (3.26)	26.9 (3.19)	0.39
	cdr	0 to 1	0.237 (0.327)	0.306 (0.334)	0.56
MRI, from the CSV	etiv	1,123 to 1,987	1,494 (179)	1,468 (145)	0.51
	nwbv	0.66 to 0.837	0.735 (0.0366)	0.729 (0.0289)	0.46
	asf	0.883 to 1.56	1.19 (0.14)	1.21 (0.118)	0.49
Timing	days_since_baseline	0 to 2,288	308 (500)	79.5 (217)	0.39
	days_to_next	182 to 1,707	709 (248)	780 (339)	0.51
	has_prev_visit	0 to 1	0.359 (0.481)	0.129 (0.341)	0.40
	days_since_prev	0 to 1,554	244 (362)	79.5 (217)	0.40
Change, looking back	mmse_change_prev	−12 to 5	−0.0521 (1.38)	0 (0.258)	0.49
	nwbv_change_prev	−0.034 to 0.011	−0.00263 (0.00623)	−0.000581 (0.00258)	0.60
	cdr_change_prev	−0.5 to 0.5	0.0156 (0.113)	0 (0)	0.49
	mmse_rate_prev	−13.6 to 3.26	−0.103 (1.33)	−0.00851 (0.162)	0.49
	nwbv_rate_prev	−0.0213 to 0.00891	−0.00147 (0.0036)	−0.000282 (0.00124)	0.60
From the raw scan	mri_csf_proxy	0.17 to 0.369	0.261 (0.0406)	0.255 (0.0392)	0.48

Among declined pairs, both time-since columns share a mean of 79.5, because so many of those rows are first visits

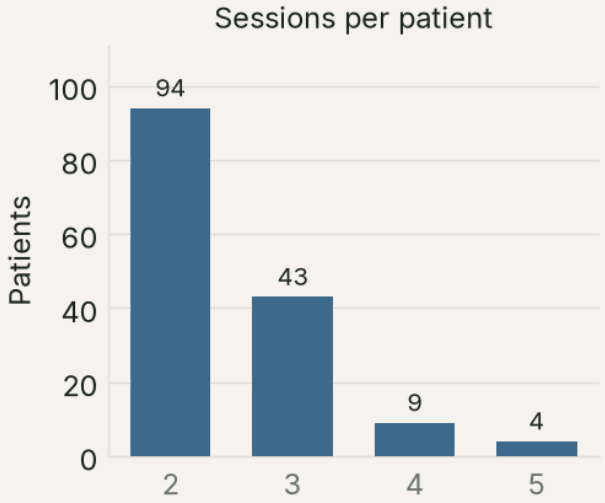
INVENTORY

Every volume is the same shape

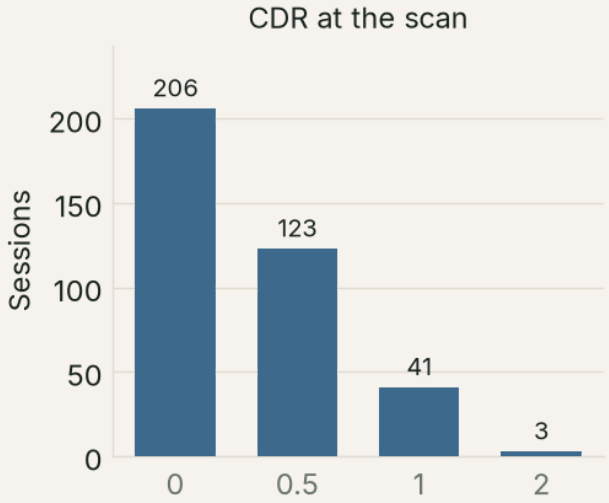
That is the quiet gift in this archive. All 1,367 scans are T1-weighted MPRAGE, sagittal, Analyze 7.5, big-endian float32, on a grid of 256 × 256 × 128 at 1 × 1 × 1.25 mm. They can be stacked before anyone resamples them.



Most sessions were scanned four times. 249 sessions have four repeats, 123 have three, and one session has two. 1×2 + 123×3 + 249×4 = 1,367 scans.



Follow-up is short. 94 patients have two sessions, 43 have three, 9 have four, and 4 have five. That is why 150 people produce 223 pairs and no more.



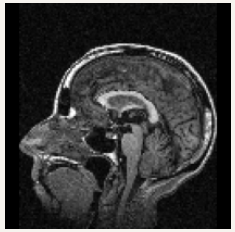
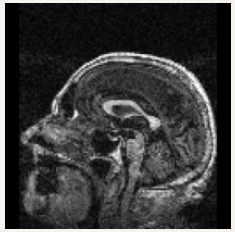
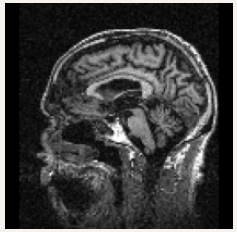
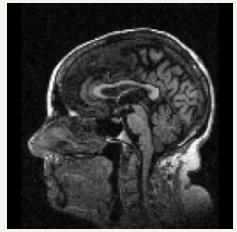
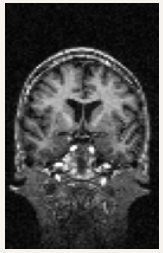
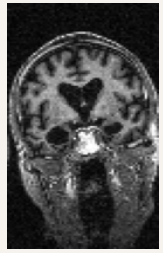
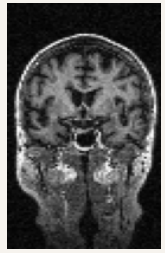
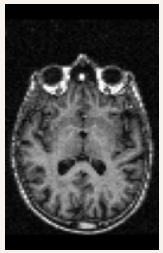
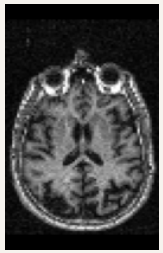
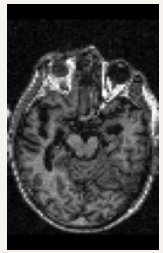
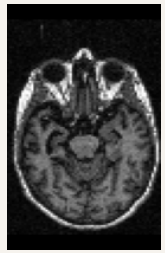
Most scans are of people at CDR 0. 206 sessions at 0, 123 at 0.5, 41 at 1, and 3 at CDR 2. The pictures, like the pairs, are mostly of people who are still well.

373 sessions, one per clinical visit · 2 to 4 repeats in a session

FOUR SESSIONS, ONE PER CDR

What a raw slice actually looks like

Central cuts of the first scan in one session at each rating. Sagittal, then coronal, then axial at the level of the eyes. Skull, face, and neck are still in the frame. Nothing here has been aligned to a template.

	CDR 0 OAS2_0056_MR2 age 73 · nWBV 0.739	CDR 0.5 OAS2_0040_MR2 age 86 · nWBV 0.724	CDR 1 OAS2_0014_MR2 age 77 · nWBV 0.696	CDR 2 OAS2_0120_MR2 age 78 · nWBV 0.700
SAGITTAL				
CORONAL				
AXIAL				

On these four, nWBV steps down as CDR rises, from 0.739 to about 0.70, and the ventricles look a little roomier. Four faces can suggest that. They cannot prove it. The whole-sample overlap, on the next page and the one before, is the result that counts.

QUALITY, AND ONE NUMBER WE COMPUTED OURSELVES

The pictures are usable, and still coarse

Intensity is in scanner units

Foreground intensity varies from scan to scan, with a coefficient of variation of 19%. A CNN cannot be handed those numbers raw. Each volume needs its own scaling, a z-score inside the brain is the natural one, before the network ever sees it.

Quality flagged 19 sessions, all for motion

SNR runs from 6.7 to 36.4, mean 21.4. The SNR rule, more than three interquartile ranges below the first quartile, flagged nobody. Repeat-scan agreement is a different story. When the first scan and its repeats disagree, the person likely moved. That rule flagged 19 of 373 sessions. They stay in the file, marked with `mri_qc_flag`, so the final models can be run again without them.

> SNR at the visit has a training AUC of 0.46. A quality measure that separated the classes would have been a leak. This one does not.

The CSF proxy is a stand-in, not a segmentation

For each scan we measure the share of dark, fluid-like voxels in the central 40% of the volume, then average the repeats. It is an exploratory atrophy proxy. Across the 373 sessions it tracks nWBV at r = −0.54. On the 223 pairs the correlation is −0.52, and it is higher when CDR is higher.

It does not tell us who declines next. Mean CSF proxy is 0.261 for stable pairs and 0.255 for declined pairs. The training AUC is 0.48. For comparison, nWBV from the CSV has a training AUC of 0.46. Both describe the brain we can already see. Neither separates the future.

That is the opening the imaging model is for. Ventricle shape, the hippocampus, the pattern across a slice: none of that is in a single fraction. The CNN is being asked for structure this proxy cannot hold. The scans still include skull and face, they are not registered, and their intensity scales differ. Skull-stripping, template registration, and a common intensity scale belong to the modeling sprint, not to this file.

mri_n_scans, mri_snr, mri_repeat_r, and mri_qc_flag are audit columns, not model inputs

FINDINGS, THEN THE THREE MODELS

What the table showed, and what the scans added

THE FINDINGS

1. **The sample is small and lopsided.** 150 patients, 223 pairs, 1,367 scans, and every pair has pictures at both visits. 31 pairs declined. The test set holds 6 of them.
2. **Cognition is the clearest clinical signal.** Mean MMSE is 26.9 where the next visit declined and 27.9 where it did not. The decline rate is 19% at CDR 0.5 and 11% at CDR 0.
3. **Most of the decline is early.** About 18% of first-visit pairs, against about 5% of later pairs. The change features are 0 on all 150 first visits.
4. **Brain volume, measured either way, is weak alone.** nWBV separates at 0.54. The CSF proxy tracks it and reaches only 0.52. eTIV and ASF are the same measurement (r = −0.99).
5. **Scan quality is not a hidden copy of the label.** SNR's training AUC is 0.46. Nineteen sessions are flagged for motion and kept for a sensitivity run.

HOW THIS SHAPES SPRINT 3

1. **Score the rare class.** A prediction of "stable" for every row is right on 86% of them and finds none of the declines. We report recall, F1-score, and ROC-AUC, with class weights or resampling.
2. **Choose models on the folds.** Five patient-grouped folds inside training. The test set is touched once, and its metrics are reported with that thinness written beside them.
3. **Metadata-only.** The 19 clinical columns. A linear model keeps either eTIV or ASF, and it sees `has_prev_visit` beside the change features so "no history" is not read as "no change."
4. **MRI-only.** A CNN on the preprocessed scans: average the two to four repeats, skull-strip, register to a template, z-score intensities, and resample, for example to 128^3 or to 2D slices. With 223 pairs, a small network, augmentation, early stopping, or transfer learning.
5. **Both together.** The CNN embedding joined to the metadata, on the same patient-level splits, so the three models can be compared. Run the final ones again without the 19 flagged sessions.

The horizon has a name. Days to the next visit run from 182 to 1,707. That column is the window the label refers to, and it has to be known when the prediction is made. Volume also needs age beside it: nWBV falls as age rises (r = −0.51), so a coefficient on volume alone is partly a coefficient on age.

VERSION 2.0 · FOR THE NEXT PERSON WHO OPENS THE FILE

What this dataset is, and what it is not

An ML-ready longitudinal table for predicting next-visit cognitive decline. Features describe the person and their scan at the earlier visit. `declined` is 1 when CDR increases. Every row links to the raw scans of both visits through `mri_id`.

Instances	223 visit pairs from 150 patients
Features	20 model inputs: 19 clinical columns and 1 scan measure
Audit only	Scan count, SNR, repeat-scan agreement, QC flag
Target	31 positives, 13.9%. No patient declined more than once.
Splits	Train 178 pairs / 119 patients. Test 45 / 31. Five grouped folds in train. Test rows are fold −1.
Linked MRI	1,367 raw T1 scans in 373 sessions. All 223 pairs have scans at both visits.
Files	`oasis2_visit_pairs.csv`, `oasis2_mri_sessions.csv`, `oasis2_mri_scans.csv`, plus a metadata JSON for each build
Curators	Christina Dames, Cinthia Ochoa Torre, Rayna Alexandre-Petrie

What it is for

Comparing three approaches to the same question: the scans, the patient metadata, and the two together. The readers are this team and the course instructors. The work is academic.

What it is not for

Diagnosis, treatment, or any decision about a real person. The sample is small, from one research site, and it is not a picture of the general population.

Who is in it

150 patients, 41% men, ages 60 to 96 at the earlier visit (mean 76.1), all right-handed. Earlier-visit CDR: 133 at 0, 70 at 0.5, 20 at 1. Someone with k visits contributes k − 1 pairs.

Where it came from

OASIS-2, Marcus et al. 2010, Washington University Knight ADRC. The clinical table is the Kaggle export, used unmodified. The scans are the two official archives from oasis-brains.org. `make_mri_dataset.py` streams them. `make_dataset.py` cleans, pairs, attaches the session measures, splits, and imputes.

OASIS2_VISIT_PAIRS.CSV

What a row contains

Identifiers, kept out of the model

subject_id	Patient. The grouping key.
mri_id	MRI session of the earlier visit. Joins to the MRI tables.
visit	Visit number of the earlier visit.

The person

age	Years at the earlier visit.
sex_male	1 = male, 0 = female.
educ	Years of education.
ses	Socioeconomic status, 1 highest to 5 lowest.
ses_imputed	1 if SES was missing and filled in.

Scores and the MRI summaries from the CSV

mmse	Mini-Mental State Examination, 0 to 30.
cdr	Clinical Dementia Rating now: 0, 0.5, or 1.
etiv	Estimated intracranial volume, cm^3.
nwbv	Whole-brain volume as a fraction of eTIV.
asf	Atlas scaling factor, about 1 / eTIV.

Time, and change since the previous visit

days_since_baseline	Days since this patient's first visit.
days_to_next	Days until the next visit. The horizon.
has_prev_visit	1 if this visit itself has a visit before it.
days_since_prev	Days since the previous visit. 0 if none.
mmse_change_prev	MMSE change. 0 if there is no previous visit.
nwbv_change_prev	nWBV change. 0 if none.
cdr_change_prev	CDR change. 0 if none.

Time, continued

mmse_rate_prev	MMSE change per year. 0 if none.
nwbv_rate_prev	nWBV change per year. 0 if none.

From the raw scan

mri_csf_proxy	Share of dark, CSF-like voxels in the central 40% of the scan, averaged over repeats. A model input. r = −0.54 with nWBV on the 373 sessions.
mri_n_scans	Repeats in the session, 2 to 4. Audit only.
mri_snr	Mean foreground divided by background SD. Audit only.
mri_repeat_r	Pearson r between the first scan and each repeat. Low means motion. Audit only.
mri_qc_flag	1 if SNR or repeat agreement is more than 3 IQR below Q1. Audit only.

The label, and where the row sits

mmse_imputed	1 if MMSE was filled from the previous visit. Audit only.
cdr_next	CDR at the next visit. Defines the target. Never a feature.
declined	The target. 1 if cdr_next is greater than cdr.
split	train or test, assigned by patient.
cv_fold	0 to 4 for training rows. −1 for test rows.

The two MRI tables

sessions	`oasis2_mri_sessions.csv`, 373 rows. The same measures, once per session, plus header fields.
scans	`oasis2_mri_scans.csv`, 1,367 rows. The same measures once per scan, plus intensity percentiles.

20 of these columns are model inputs · identifiers, audit columns, and cdr_next are not

HOW THE FILE WAS MADE, AND WHERE IT BENDS

Rebuild order, limits, and the citation

CLINICAL BUILD, IN ORDER

1. Validate the schema and check that each MRI ID is unique.
2. MMSE: 2 missing visits, 1 patient, filled from the previous score. One pair is affected.
3. Change features look backward only.
4. Pair each visit with the next. 373 visits become 223 pairs.
5. Attach the MRI session measures of visit t. All 223 match.
6. StratifiedGroupKFold holds out about 20% of patients. Training is split into 5 grouped folds.
7. SES for 8 patients, 11 pairs, filled with the training mode, SES = 2.
8. Drop Group, which leaks, and Hand, which is constant.
9. Encode M/F as `sex_male`. Names to snake_case.

MRI BUILD

Stream each archive and decode with nibabel. Nothing is extracted. Per scan: header, intensity percentiles above an Otsu threshold, SNR, CSF proxy. Per session: average the metrics, measure repeat agreement on 4×-downsampled volumes, add the QC flag. Half-resolution central slices of the first scan per session are kept under data/interim for figures, and are not versioned. Skull-stripping, registration, and intensity normalization wait for the CNN.

LIMITS WORTH SAYING OUT LOUD

223 pairs and 31 positives, 6 of them in test. Choose models on the folds. Accuracy flatters a constant “stable” prediction, so report recall, F1, and ROC-AUC. Visit gaps run from 182 to 1,707 days. CDR moves in steps of 0.5. Five pairs improve and are labeled stable. Only three reach CDR 2. Change features are 0 on 150 first visits. eTIV and ASF repeat each other. The raw scans include skull and face, are not registered, and do not share an intensity scale. The CSF proxy is coarse, not a validated segmentation. One site, right-handed volunteers aged 60 and over, no race or ethnicity field. A result here may not travel.

Ethics. The data are de-identified. We do not try to re-identify anyone or link them elsewhere. Use follows the OASIS agreement and the acknowledgment rules at oasis-brains.org. Raw scans stay out of the repository. A prediction from this file is a research output. It is not medical advice.

Rebuild, in this order. `python -m src.data.make_mri_dataset` (about 2 to 5 minutes, both archives in data/raw), then `python -m src.data.make_dataset`, then `python -m src.visualization.eda_report`. If a hash, a row count, an imputation value, or a split statistic changes, the dataset version changes with it.

Citation. Marcus, D. S., Fotenos, A. F., Csernansky, J. G., Morris, J. C., & Buckner, R. L. (2010). Open Access Series of Imaging Studies: Longitudinal MRI data in nondemented and demented older adults. *Journal of Cognitive Neuroscience, 22*(12), 2677–2684.